

Image-Text Retrieval With Cross-Modal Semantic Importance Consistency

Zejun Liu¹, Fanglin Chen¹, *Member, IEEE*, Jun Xu², *Member, IEEE*, Wenjie Pei¹,
and Guangming Lu¹, *Senior Member, IEEE*

Abstract—Cross-modal image-text retrieval is an important area of Vision-and-Language task that models the similarity of image-text pairs by embedding features into a shared space for alignment. To bridge the heterogeneous gap between the two modalities, current approaches achieve inter-modal alignment and intra-modal semantic relationship modeling through complex weighted combinations between items. In the intra-modal association and inter-modal interaction processes, the higher-weight items have a higher contribution to the global semantics. However, the same item always produces different contributions in the two processes, since most traditional approaches only focus on the alignment. This usually results in semantic changes and misalignment. To address this issue, this paper proposes Cross-modal Semantic Importance Consistency (CSIC) which achieves invariance in the semantic of items during aligning. The proposed technique measures the semantic importance of items obtained from intra-modal and inter-modal self-attention and learns a more reasonable representation vector by inter-calibrating the importance distribution to improve performance. We conducted extensive experiments on the Flickr30K and MS COCO datasets. The results show that our approach can significantly improve retrieval performance, proving the proposed approach's superiority and rationality.

Index Terms—Cross-modal, image-text retrieval, self-attention, semantic importance, alignment.

I. INTRODUCTION

WITH the explosive growth of social media and Internet multimedia data, image-text retrieval has become a research hotspot in academic and industry [1], [2], [3], [4],

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Zejun Liu, Fanglin Chen, and Guangming Lu are with the Harbin Institute of Technology at Shenzhen, Shenzhen 518055, China, and also with the Guangdong Provincial Key Laboratory of Novel Security Intelligence Technologies, Shenzhen 518055, China (e-mail: 20s151105@hit.edu.cn; chenfanglin@hit.edu.cn; luguangm@hit.edu.cn).

Jun Xu and Wenjie Pei are with the Harbin Institute of Technology at Shenzhen, Shenzhen 518055, China (e-mail: xujunqy@hit.edu.cn; wenjiecoder@outlook.com).

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[5], [6], [7], [8], [9]. As an important downstream task of cross-modal representation learning, the goal of image-text retrieval is to construct a bidirectional retrieval process for images and texts, i.e., given a query sample of a modality, retrieve semantically similar samples from the opposite modality. The task presents significant challenges due to the semantic differences and data heterogeneity between vision and language. It is necessary not only to understand the semantic information of the two modalities but also to explore the deeper correspondence.

To bridge this gap, a straightforward idea is to embed the global features of images and texts into a shared space by having the embedding vectors of semantically similar sample pairs close to each other and modeling the similarity of pairs using correlation measures. Earlier works [10], [11], [12] have learned complex transformation functions for two or more kinds of data by unsupervised methods, most of which follow a nonlinear extension of the linear method canonical correlation analysis (CCA). Supervised methods [13], [14], [15], [16] use label information to learn more discriminative vector representations. These methods force the representation vectors of matched sample pairs to be close to each other in the embedding space, while the representation vectors of non-matched sample pairs are far from each other. For example, DSCMR [15] maps the image-text representation vectors into a common space through multiple fully-connected layers, and applies the loss function to drive the representation vectors of matched pairs close. However, such works only explore the coarse-grained alignment of the two modalities and do not reveal the fine-grained correspondence of image-text pairs.

Thanks to the development of deep learning, methods to explore the fine-grained potential correspondence of cross-modal images and texts have been proposed [17], [18], [19], [20], [21]. These methods calculate the global similarity of image-text by detecting image regions at the object level and coarsely computing the similarity of all regions and words on this basis. Karpathy et al. [19] extracted items features of images and texts to build dense alignments of fine-grained information and used the average value of maximum region-word similarity score to all words as the global similarity. However, the effect of visual context on words is not considered. Since most models performing fixed-step attentional reasoning [22], [23] only focus on the limited number of semantic alignments, Lee et al. [24] proposed a stacked cross-attention that discovers all possible

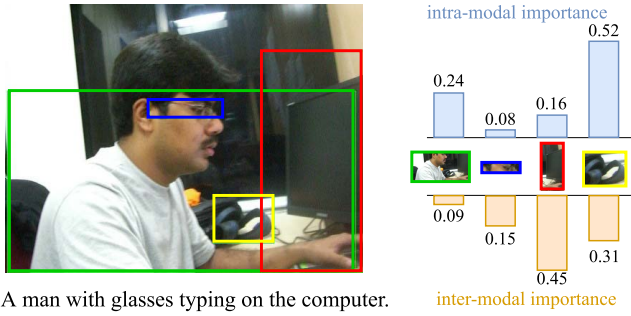


Fig. 1. An example of previous approaches do not achieve cross-modal importance consistency in visual modality. Comparing the upper (intra-modal) and lower (inter-modal) importance distributions after aligning, the regions ‘man’, ‘glasses’ and ‘computer’ with the importance weight (0.24, 0.08 and 0.16) within the modality received importance weights (0.09, 0.15 and 0.45) between the modalities, which leads to incorrect semantic relations.

alignments simultaneously to establish a more comprehensive correspondence.

Despite that these fine-grained alignment methods obtain more reasonable alignment relationships by weighting the combination of different items, they lack attention on semantic importance consistency of items. Each item has a unique embedding vector representation, and the position marked by this vector is the semantic information of the data item in the embedding space. Different items will receive different weights when they are associated within and between modalities. Some items with higher weights are more important in the global semantics, otherwise, they are less important. In general, aligning image-text describing the same scene presupposes understanding the global semantic information of image and text, i.e., the semantic association of items within a modality. This semantic association will highlight the items that are more important in global semantics. In addition, these important items can often find their counterparts in another modality, and they have a higher contribution to the alignment of shared semantics. Cross-modal alignment is achieved by fusing and associating the items of two modalities with each other. On the other hand, in the ideal alignment state, the representation vectors of matched sample pairs are close to each other and express similar semantics. In this case, the importance obtained by interacting with items from different modalities should be nearly identical, i.e., the intra-modal semantic importance is consistent with the inter-modal semantic importance. However, previous studies usually strive for achieving more accurate alignment with little or no attention to the invariance of the importance of items in the global semantics, which in turn affects the performance of the model.

Modeling without semantic importance consistency may cause the representation vector to focus on irrelevant items, resulting in alignment bias. A traditional alignment example is shown in Fig. 1, given matched image-text pairs, the intra-modal (within modality) and inter-modal (between modality) importance weights of the visual regions after aligning are in disagreement. Specifically, for the image regions ‘man’, ‘glasses’, and ‘computer’ in the visual modality, their intra-modal semantic importance (0.24, 0.18 and 0.16) and

inter-modal semantic importance (0.09, 0.15 and 0.45) are not consistent. This leads to semantic changes in the visual modality and can impair the joint representation with multimodal information. In addition, as important components of visual semantics, region ‘human’ have similar intra-modal importance to region ‘computer’ (0.24 and 0.16), yet there is a huge difference in inter-modal semantic importance (0.09 and 0.45), which indicates that region ‘human’ does not contribute to the alignment of shared semantics as much as it should. This can mislead the representation learning of the model, reducing the similarity of the positive image-text pairs and ultimately affecting the accuracy of the model. In practice, researchers usually perform region detection by pre-trained Faster R-CNN [25] and used the results as the object of the alignment. Due to oversampling, too many visual features are selected to learn cross-modal representation, which can make this situation more common and serious.

Inspired by the above observations, this paper proposes Cross-modal Semantic Importance Consistency (CSIC) for image-text retrieval, which aims to learn image-text representation with consistent semantic relations within and between modalities when performing image-text cross-modal interaction. During the training process, in addition to evaluating the semantic similarity of the image-text pairs (i.e., the representation vectors of positives are close to each other in the embedding space and the representation vectors of negatives are far from each other), further propose a scheme to make the semantic importance of the image-text items to be consistent when establishing semantic connections with items of different modalities. By imposing such a consistency constraint on the semantic importance, the model can learn more accurate representation vectors using the semantic information within and between modalities complementarily. It also promotes the representation of the positives to be further approached in the embedding space to obtain higher semantic similarity.

Specifically, we characterize the semantic relationships among items by processing the self-attention matrix and design the semantic importance measure. After that, we process all items of image-text to obtain the intra-modal and inter-modal semantic importance distributions, i.e., the normalized importance weights for all items of image and text, and drive the two distributions consistently. We conduct relevant experiments to verify the effectiveness of the proposed method. The two importance distributions of items within and between modalities are calibrated by CSIC to better learn the representation vectors of image-text pairs. The effective improvement of the model performance of image-text retrieval on the Flickr30K and MS COCO datasets demonstrates the superiority of the proposed scheme.

The contributions of our work are summarized as follows:

- For image-text retrieval, we propose semantic importance consistency to ensure the semantic consistency of image-text interactions across modalities. The scheme simultaneously exploits intra- and inter-modal semantic links, enabling the model to learn more reasonable and robust representation vectors.
- We propose a semantic importance calculation method based on the attention mechanism and design the loss

function so that the two importance distributions tend to be consistent during the training process.

- Two versions of UNITER are used as backbone separately and extensive experiments are conducted on two benchmark datasets, i.e., Flickr30K and MS COCO. And we conducted a rich qualitative analysis on the Flickr30K. The experimental results demonstrate the effectiveness of our technique.

The rest of this paper is organized as follows. In Section II, we briefly summarize some prior works on cross-modal image-text retrieval. In Section III, we present the details of the proposed CSIC and the loss function. In addition, experimental results and analysis are discussed in Section IV, where we perform a quantitative comparison of CSIC with previous models, and the experimental results show the superiority of the proposed method. Finally, we conclude this work in Section V.

II. RELATED WORK

Image-text retrieval is a key research problem in the Vision-and-Language task, the essence of which is to explore the potential semantic correspondence of vision and language. It can be roughly categorized into two groups: 1) binary representation learning; 2) real-valued representation learning. Binary representation learning maps the features of two modalities into Hamming space for retrieval, which is computationally efficient and faster but brings a decrease in retrieval accuracy. Real-valued representation learning can be subdivided into coarse-grained and fine-grained alignments, which explore semantic correspondences using global or local features of image and text, respectively.

A. Coarse-Grained Alignment

The coarse-grained alignment approach aims to explore the semantic similarity of matched pairs by mapping the global features of images and texts into a shared embedding space. Images are encoded using deep convolutional neural networks, and recurrent neural networks are used to encode texts. A large number of studies have explored image-text global features for matching in the embedding space [15], [26], [27], [28], [29], [30], [31], [32]. Zhen et al. [15] proposed a method that maps samples of different modalities into a common representation space and learned invariant features of the samples, enabling direct comparison of sample similarities in that space. Kiros et al. [26] used deep convolutional networks and recurrent neural networks to encode images and texts, respectively, and trained model using fold-based triplet ranking loss for the first time. Wang et al. [28] proposed a two-branch neural network to learn the joint embedding vectors of images and texts, combining cross-view ranking constraints and intra-attempt neighborhood structure constraints. Faghri et al. [31] used the triplet loss focusing on hard negatives to train models and obtained significant improvements. In addition, due to the powerful learning ability of Generative Adversarial Networks (GAN) [33] in learning discriminative representations, researchers have successively proposed some adversarial learning approaches [14], [34]. Gu et al. [32] proposed a model

that uses RNN and CNN to encode texts and images respectively, after which image features are decoded to generate text sentences, and the similarity between the two texts is measured using BLEU. And text features are treated as noise in the GAN to influence the image generation.

B. Fine-Grained Alignment

Recently, the focus of research has tended to explore fine-grained alignment relations of cross-modal image-text retrieval [19], [24], [35], [36], [37], [38], [39]. Karpathy et al. [19] used R-CNN [40] to detect salient object regions in images and used these region features as inputs to a model to explore complex interaction information and infer the semantic similarity of images and texts. Due to the features of the attention mechanism [41], it was introduced into the field of image-text retrieval and has greatly advanced the field. Lee et al. [24] proposed a stacked cross-attention approach that discovers fine-grained alignments through two symmetric attention computational structures and uses an ensemble model to achieve optimal results. Chen et al. [37] acquired alignment relations between images and texts by overlaying multiple attention mechanism and refining the alignment knowledge each time by memory distillation units. Chen et al. [42] used recurrent neural networks to symmetrically process images and sentences and reorder image objects based on the position of the most relevant words in the text to establish connections between semantically related objects. RRTC [35] constructs fine-grained correspondences with intra-modal semantic links by modeling the connections between regions and reassigning region-word similarity. Also, the global semantics of the images are summarized by the topic constraint module. Qu et al. [36] developed a dynamic multimodal interaction network with routing mechanism and explored different levels of modal interactions using four types of units. Each unit integrates dynamic routers that can activate different paths for different inputs. Wen et al. [38] proposed a cross-memory network with pairwise discrimination. The common knowledge between different modalities is captured through cross-memory while distinguishing category labels for pairwise discrimination. Ma et al. [39] proposed a network that generates image-text embeddings using multilayer nonlinear mappings based on multiple representations of images and sentences and further explores the relationship between each modality.

There are other interesting works on the idea of using fine-grained features to explore semantic correspondences. Some approaches [43], [44], [45] perform information filtering based on region-word features to learn features representing global semantics. ADAPT [43] improves the embedding representation of another modality by anchoring the global information of the modality, using a filtering strategy to modify intermediate features to construct a better embedding vector. To explore intra-modal associations of fine-grained information, some approaches [46], [47], [48] use Graph Convolutional Network (GCN) [49] to enhance the concept of global semantics of the representation vector. VSRN [47] establishes connections between image regions and performs reasoning through GCN

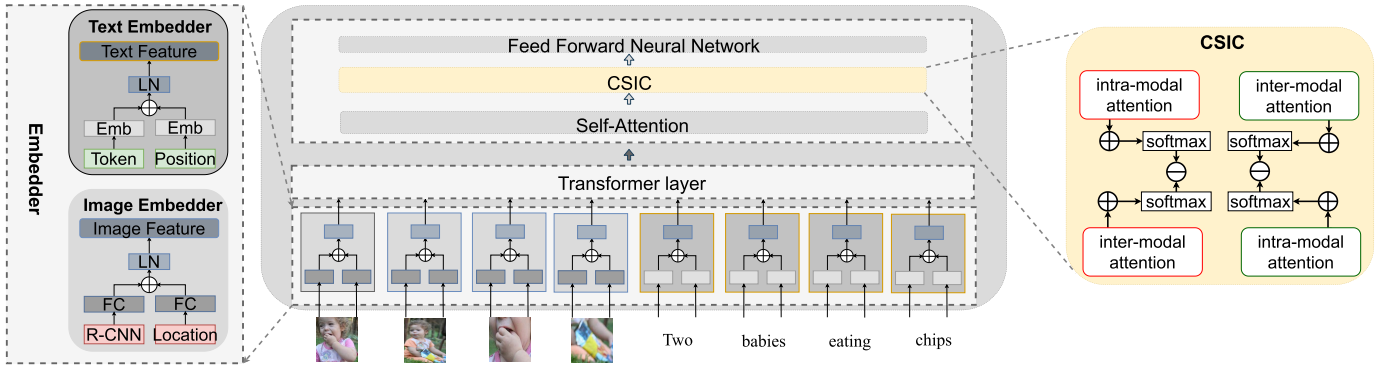


Fig. 2. An overview of the proposed CSIC. It consists of an image embedder, text embedder and transformer module with UNITER as the backbone. CSIC is applied after the self-attention in the transformer layer.

to generate features with semantic relationships. Then global semantic reasoning is performed on these enhanced features with gates and memory mechanism.

Large-scale pre-trained models are also widely used in cross-modal tasks. Most of the pre-trained models use transformer module to learn the contextual embedding of each region and word. These approaches can be categorized as dual-stream models [50], [51], [52] and single-stream models [53], [54], [55], [56], [57]. Dual-stream models use two independent transformers to learn the intra-modal contextual representation of image and text separately, and then augment the cross-modal association by a single transformer. The single-stream models, on the other hand, learn the joint contextual representation of image and text through a single transformer. In addition, Andonian et al. [58] proposed a pre-training framework using progressive self-distillation and soft image-text alignment, which facilitates learning from noisy data to robust image-text representations. Yang et al. [59] proposed triple contrast learning for vision-language pre-training using multimodal information self-supervision, which preserves semantic similarity within data modalities. ViSTA [60] uses transformer to embed image patches and text to learn an aggregated visual representation for cross-modal retrieval.

III. OUR APPROACH

In this section, we present our proposed CSIC for cross-modal image-text retrieval in detail. Fig. 2 shows the framework of the proposed method, which uses UNITER [53] as the backbone network. UNITER inputs visual and linguistic items together into transformer encoder, modeling the multimodal attention and providing a basis for measuring the intra-modal and inter-modal semantic importance of items. Given an image-text pair, we first apply the image embedder to process the visual and positional features of the image items and get the image embedding. Similarly, we use the text embedder to get the embeddings of the text items. Then the embeddings of these items are input to a transformer for representation learning, where the transformer is composed of multiple self-attention layers. Finally, we apply the proposed CSIC after the highest self-attention layer to achieve semantic importance consistency of the image and text. Compared with SCAN, CSIC uses intra-modal and inter-modal attention matrices to

measure the semantic importance of items and enables the model to learn semantically consistent vector representations by keeping the two importance distribution of samples similar.

This section is structured as follows: We first briefly introduce the backbone network, a representative model of single-stream structure. After that we show the computational mechanism of attention for multiple modalities and use the self-attention matrix to construct semantic relations between items within and between modalities. Then we propose a measure of the intra-modal and inter-modal semantic importance of items. Afterward, we elaborate on the consistency constraint of the two importance distributions. Finally, we give the loss function for training.

A. Model Overview

Given a pair of image and text, the visual regions and the textual words are used as the input to the model. The visual and textual embeddings are extracted separately by two embedders. After that, these embeddings are fed into a multi-layer transformer module to learn the joint embedding representation of region and text.

1) *Image Embedder*: In the image embedder, the visual features of each region are first extracted using Faster R-CNN, while the location features are encoded with normalization. Then the two features are mapped into the same embedding space through fully-connected (FC) layers and the outputs are summed. Finally, the embedding features of the region are obtained through layer normalization (LN).

2) *Text Embedder*: We first tokenize the text and sum the word embedding and positional embedding for each sub-word token. Then the final embedding representation of the text is obtained via another LN layer.

After that, the visual features and text features are fused by a single-transformer module to obtain the final joint features. CSIC is applied to the self-attention layer in the transformer to facilitate the alignment of multimodal information. Our approach can be constrained for any layer of self-attention and can also be applied to multiple layers.

B. Attention Submatrix Extraction

In this subsection, we describe the extraction of intra-model and inter-model attention matrix based on UNITER.

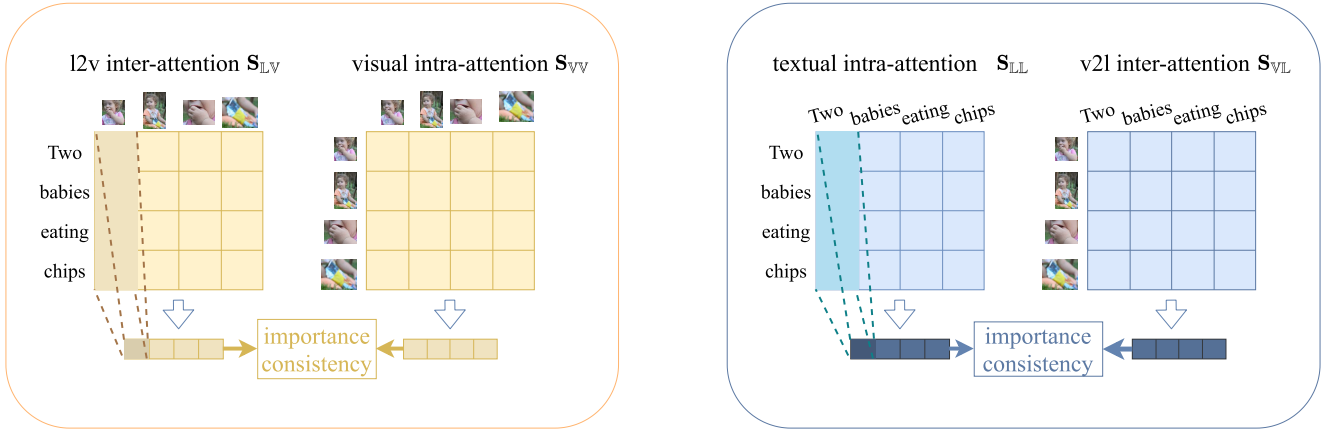


Fig. 3. Cross-modal semantic importance congruence. The left and right sides are the semantic importance consistency of visual modality and text modality, respectively. Only some representative items of text and image are selected for illustration.

It is mainly composed of a multilayer transformer layers with self-attention mechanism to learn cross-modal contextual representation for images and texts. Given $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times d_k}$, the matrices of N query, key and value vectors of dimension d_k , the scaled dot-product attention matrix is computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} = \sigma(\mathbf{S})\mathbf{V}, \quad (1)$$

where $\mathbf{S}$ is the attention score matrix and S_{ij} measures the similarity of the i_{th} query vector to the j_{th} value vector. For the convenience of representation, we use $\mathbb{V}$ for visual modality and $\mathbb{L}$ for textual modality. Given the embedding vectors of matched pairs, i.e., the embedding vector $\mathbf{X}_V = \{v_i \mid i = 0, \dots, N_V, v_i \subseteq \mathbb{R}^d\}$ for image I and $\mathbf{X}_L = \{t_i \mid i = 0, \dots, N_L, t_i \subseteq \mathbb{R}^d\}$ for text T . Connect $\mathbf{X}_V$ and $\mathbf{X}_L$ to obtain $\mathbf{X} = [\mathbf{X}_V \mid \mathbf{X}_L]$ with length $N = N_V + N_L$, which is the input to CSIC. Then the query and key vectors can be represented as:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}^Q = \begin{pmatrix} \mathbf{X}_V \\ \mathbf{X}_L \end{pmatrix} \mathbf{W}^Q = \begin{pmatrix} \mathbf{Q}_V \\ \mathbf{Q}_L \end{pmatrix}, \quad (2)$$

$$\mathbf{K} = \mathbf{X}\mathbf{W}^K = \begin{pmatrix} \mathbf{X}_V \\ \mathbf{X}_L \end{pmatrix} \mathbf{W}^K = \begin{pmatrix} \mathbf{K}_V \\ \mathbf{K}_L \end{pmatrix}, \quad (3)$$

where $\mathbf{W}^Q$ and $\mathbf{W}^K$ are to-be-learned weight parameter. Correspondingly, the attention score matrix $\mathbf{S}$ can be divided according to the rules of matrix operations as:

$$\begin{aligned} \mathbf{S} &= \mathbf{Q}\mathbf{K}^T = \begin{pmatrix} \mathbf{Q}_V \\ \mathbf{Q}_L \end{pmatrix} \begin{pmatrix} \mathbf{K}_V^T & \mathbf{K}_L^T \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{Q}_V\mathbf{K}_V^T & \mathbf{Q}_V\mathbf{K}_L^T \\ \mathbf{Q}_L\mathbf{K}_V^T & \mathbf{Q}_L\mathbf{K}_L^T \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{S}_{VV} & \mathbf{S}_{VL} \\ \mathbf{S}_{LV} & \mathbf{S}_{LL} \end{pmatrix}. \end{aligned} \quad (4)$$

The diagonal submatrices $\mathbf{S}_{VV}, \mathbf{S}_{LL}$ represent the intra-modal attention scores of text modality and visual modality, respectively. The $\mathbf{S}_{LV}, \mathbf{S}_{VL}$ on the negative diagonal represent the inter-modal attention scores of text items to visual items and the opposite.

C. Semantic Importance

The detailed treatment of cross-modal semantic importance consistency is shown in Fig. 3. We consider the four normalized submatrices i.e., applying the row-wise scaled softmax function separately, as the intra-modal and inter-modal semantic relations of items. For notational simplicity, the normalized submatrices will use the same representation as before normalization. Since the self-attention mechanism is good at capturing the internal relevance of data or features, S_{ij} measures the interplay between visual and textual items. We consider the self-attention matrix $\mathbf{S}_{LL}$ as the semantic linkage between textual items, each row represents the relationship between an item and other textual items. $\mathbf{S}_{LV}$ is the semantic linkage between textual items and visual items, each row represents the relationship between a textual item and all visual items. $\mathbf{S}_{VV}, \mathbf{S}_{VL}$ are the same. We use $\mathbf{S}_{LL}$ and $\mathbf{S}_{VL}$ to construct the intra-modal and inter-modal semantic importance of textual items, while $\mathbf{S}_{VV}$ and $\mathbf{S}_{LV}$ are used to construct the intra-modal and inter-modal semantic importance of visual items.

1) *Image Semantic Importance*: Image semantic importance consistency attempts to ensure that the semantic of visual items remain constant while learning the embedding features. The attention scores reflect the weight of the visual items in the context vector of items in the row dimension, and a higher score indicates greater semantic importance. The semantic importance weight distribution of the visual items is obtained by summing the attention matrix over the column dimension and normalizing it.

Specifically, given feature set $\mathbf{X}_V$ of I , the intra-modal self-attention matrix of the image is $\mathbf{S}_{VV} \in \mathbb{R}^{N_V \times N_V}$. The attention score of the i_{th} visual item v_i to the j_{th} visual item v_j can be represented as $\mathbf{S}_{VV}[i, j]$, it also measures the semantic importance of v_j in the contextual representation vector of v_i . We compute the global semantic importance of v_j in the visual modality as:

$$z_i^V = \sum_{j=1}^{N_V} \mathbf{S}_{VV}[i, j]. \quad (5)$$

Eq.(5) quantifies the global semantic importance of a visual item within a modality, and then we perform softmax

normalization on the importance scores of all items to obtain the intra-modal semantic importance distribution $A_V \in \mathbb{R}^{N_V}$ for all regions of I :

$$A_V[i] = \frac{\exp(\lambda_1 z_i^V)}{\sum_{j=1}^{N_V} \exp(\lambda_1 z_j^V)}, \quad (6)$$

where λ_1 is the inverse temperature of the softmax function.

Since embedding vectors from different modalities need to interact and achieve alignment through inter-modal self-attention, we further extract the inter-modal semantic importance of regions by means of the inter-modal attention matrix $S_{LV} \in \mathbb{R}^{N_L \times N_V}$. Specifically, $S_{LV}[i, j]$, measures the attention weight of the i_{th} textual item t_i on the j_{th} visual item v_j and also represents the semantic importance of v_j in the contextual representation vector of t_i . Then the global semantic importance of v_j when interacting with the items of language can be expressed as:

$$c_i^V = \sum_{j=1}^{N_V} S_{LV}[i, j]. \quad (7)$$

Similarly, we can obtain the inter-modal semantic importance distribution $B_V \in \mathbb{R}^{N_V}$ for all regions of I :

$$B_V[i] = \frac{\exp(\lambda_1 c_i^V)}{\sum_{j=1}^{N_V} \exp(\lambda_1 c_j^V)}, \quad (8)$$

where λ_1 is the same inverse temperature as in Eq.(6).

2) *Text Semantic Importance*: In contrast to the image semantic, text semantic importance sum the S_{LL} and S_{VL} on the column-wise and normalize to obtain the importance weight distribution of language items. In detail, given a text feature set X_L with N_L items, the intra-modal attention matrix is S_{LL} . The global semantic importance of the i_{th} text item t_i in the text modality can be computed as:

$$z_i^L = \sum_{j=1}^{N_L} S_{LL}[i, j]. \quad (9)$$

Then we can get the semantic importance distribution $A_L \in \mathbb{R}^{N_L}$ of all items:

$$A_L[i] = \frac{\exp(\lambda_2 z_i^L)}{\sum_{j=1}^{N_L} \exp(\lambda_2 z_j^L)}, \quad (10)$$

where λ_2 is the inverse temperature of the softmax function. After that, we calculate the inter-modal semantic importance distribution $B_L \in \mathbb{R}^{N_L}$ as:

$$B_L[i] = \frac{\exp(\lambda_2 c_i^L)}{\sum_{j=1}^{N_L} \exp(\lambda_2 c_j^L)}, \quad (11)$$

where λ_2 is the same inverse temperature as in Eq.(10) and $c_i^L = \sum_{j=1}^{N_V} S_{VL}[i, j]$.

D. Learning With Semantic Importance Consistency

Given a matched image-text pair, the semantic importance of their items can be seen as an essential property of image and text, this semantic importance should not differ across embedding spaces. Therefore, we propose to learn embedding vectors with consistent semantic importance, requiring each item to maintain a steady semantic state across embedding spaces.

Specifically, as described above, given an image I , two semantic importance distributions A_V and B_V denote the intra-model and inter-model semantic importance distribution of the vision items, respectively. To make the two distributions consistent, we define the dispersion L_{CSIC-V} between them as:

$$L_{CSIC-V} = KL(A_V, B_V) + KL(B_V, A_V), \quad (12)$$

where $KL(\cdot)$ denotes the Kullback-Leibler Divergence. Similarly, we can obtain the semantic importance dispersion of textual items:

$$L_{CSIC-L} = KL(A_L, B_L) + KL(B_L, A_L). \quad (13)$$

Finally, CSIC loss is defined as:

$$L_{CSIC} = L_{CSIC-V} + L_{CSIC-L}. \quad (14)$$

The detailed processing of cross-model semantic consistency is illustrated in Alg.1.

Algorithm 1 Cross-Modal Semantic Importance Consistency Algorithm

Input: self-attention matrices $S_{VV}, S_{LV}, S_{LL}, S_{VL}$

Output: L_{CSIC}

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1: for  $x, y \in \{V, L\}$  and  $x \neq y$  do
2:   for  $i = 1$  to  $N_x$  do
3:      $z_i^x \leftarrow \text{sum}(S_{xx}[:, i]);$ 
4:   end for
5:    $A_x \leftarrow \text{softmax}(z^x);$ 
6:   for  $i = 1$  to  $N_x$  do
7:      $c_i^x \leftarrow \text{sum}(S_{yx}[:, i]);$ 
8:   end for
9:    $B_x \leftarrow \text{softmax}(c^x);$ 
10:   $L_{CSIC-x} \leftarrow KL(A_x, B_x) + KL(B_x, A_x);$ 
11: end for
12:  $L_{CSIC} \leftarrow L_{CSIC-V} + L_{CSIC-L};$ 
13: return  $L_{CSIC}$ 

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E. Loss Function for Training

In order to keep the embedding vectors of matched image-text pairs close to each other in the embedding space and keep the unmatched image-text pairs away, triplet ranking loss is widely used in image-text retrieval methods. Following, we only consider the hard negatives within a mini-batch instead of considering all the negatives:

$$L_{triplet} = \sum_{j=1}^{N_n} [F(T_i, I_j) - F(T_i, I_i) + \Delta]_+ + \sum_{j=1}^{N_n} [F(T_j, I_i) - F(T_i, I_i) + \Delta]_+, \quad (15)$$

where $[x]_+ = \max(0, x)$, and $F(T, I)$ is the semantic similarity scores of image-text pair, N_n denotes the number of hard negatives. Δ is a present margin value. Image and text with same subscript is positive pair.

CSIC constraint is imposed on the attention of the last transformer layer for all positive pairs. The CSIC loss is

TABLE I

RESULTS OF IMAGE AND TEXT RETRIEVAL ON FLICKR30K AND MS COCO. R@K CORRESPONDS TO WHETHER THE GROUND TRUTH IS RECALLED IN THE TOP-K RESULTS. ‘-BASE’ DENOTES THE RESULT OF BASE VERSION OF METHODS

Model	Flickr30k							MS COCO						
	Image Retrieval			Text Retrieval			Overall	Image Retrieval			Text Retrieval			Overall
	R@1	R@5	R@10	R@1	R@5	R@10	Sum	R@1	R@5	R@10	R@1	R@5	R@10	Sum
SCO [17]	41.10	70.50	80.10	55.60	81.90	89.50	416.20	33.10	62.90	75.50	42.80	72.30	83.00	396.60
SCAN [24]	48.60	77.70	85.20	67.40	90.30	95.80	465.00	38.60	69.30	80.40	50.40	82.20	90.00	410.96
VSRR [47]	54.70	81.80	88.20	71.30	90.60	96.00	482.60	40.50	70.60	81.10	53.00	81.10	89.40	415.70
IMRAM [37]	53.90	79.40	87.20	74.10	93.00	96.60	484.20	39.70	69.10	79.80	53.70	83.20	91.00	416.50
Uncicoder-VL-base [57]	71.50	91.20	95.20	86.20	96.30	99.00	539.40	48.40	76.70	85.90	62.30	87.10	92.80	453.20
VILBERT-base [51]	58.20	84.90	91.50	-	-	-	-	-	-	-	-	-	-	-
ViLT [56]	64.40	88.70	93.08	83.50	96.70	98.60	542.98	42.70	72.90	83.10	61.50	86.30	92.70	439.20
SOHO [55]	72.50	92.70	96.10	86.50	98.10	99.30	545.20	50.60	78.00	86.70	66.40	88.20	93.80	463.70
UNITER-base [53]	72.52	92.36	96.08	85.90	97.10	98.80	542.76	50.33	78.52	87.16	64.40	87.40	93.08	460.89
CSIC-base	73.42	93.02	96.28	86.90	97.80	99.30	546.72	50.97	78.62	87.23	66.66	88.68	94.28	466.44

calculated based on the self-attention from the last transformer layer. The total training loss function is defined as:

$$L = L_{\text{triplet}} + \lambda_s L_{\text{CSIC}}, \quad (16)$$

where λ_s is a trade-off factor that dynamically balances the two losses according to the number of training iterations. Since CSIC is based on the alignment between vision and language items, it is beneficial to focus on the triplet ranking loss at the beginning of training, while rapidly increasing the weight of the CSIC loss can help the model learn a more reasonable representation vector. Therefore, we gradually increase the weights of CSIC losses using Training Signal Annealing [61], and according to [62] this, we use the following logarithmic function scheduling:

$$\lambda_s = 1 - \exp(-s/S \times 5), \quad (17)$$

where S is the total number of training step and s is the current step.

IV. EXPERIMENTS

A. Datasets and Evaluation Metric

We conduct experiments on two benchmark datasets, i.e., Flickr30k and MS COCO, and compare them with several well-known algorithms to evaluate the performance of the proposed method.

1) *Flickr30k*: Flickr30k contains 31,000 images and 158,915 English annotations from the website Flickr. Each image is annotated with 5 sentences. Here, we follow the dataset partitioning rules provided [24], [31], where the training set contains 29000 images, the validation set contains 1000 images, and the test set contains the remaining 1000 images.

2) *MS COCO*: MS COCO is a large-scale vision-linguistic task dataset containing 123k images with 5 English annotations per image. We also follow previous work [19], [24] and divide the dataset into training, validation, and test sets, containing 123287, 5000, and 5000 images, respectively.

3) *Evaluation Metrics*: We use the same evaluation metrics in this paper as previous work [24], [31], [63], namely, we use the recall at K to measure the performance of bidirectional retrieval, i.e., text retrieval given query image (text retrieval) and image retrieval given query text (image retrieval). R@K is defined as the percentage of ground-truth samples among the K results that are most similar to the query sample obtained by retrieval. [37], in order to compare the validity of the methods in a comprehensive way, we use an additional metric ‘Sum’, the summation of all evaluation metrics.

B. Implementation Details

We conduct our experiments using the base and large versions of CSIC, which are based on different sizes of backbone. We follow the original experimental settings and hyperparameter configurations. The margin in Eq.(15) is 0.2. The inverse temperature factors, i.e., λ_1 in Eq.(6) and λ_2 in Eq.(10), are set as 9 and 5 respectively. For each query sample, we randomly sample 31 hard negatives and each batch contains 32 different examples, so the batch size is 2048. The learning rate is 5e-5 and the number of training iterations is 5000. We use pytorch v1.8 and all experiments are run on 2 Geforce RTX 3090 GPUs.

C. Results on Benchmark Datasets

To verify the effectiveness of the proposed method, we conduct experiments and compare with the state-of-the-art models, and our method largely outperforms methods on different datasets. The left and right parts of Table I report the comparison results for Flickr30k and MS COCO, respectively. In addition, we list the comparison of the methods for large version in Table II.

On Flickr30k, our method can outperform both base-lines on almost all metrics. Compared to UNITER-base, our approach can achieve the largest performance improvements of 0.90%(R@1), 1.00% (R@1) for image retrieval and text retrieval, and a total performance improvement of 3.98%. Compared to the large vision, our approach achieves a maximum improvement of 1.76% (R@1) for image retrieval

TABLE II
RESULTS OF IMAGE AND TEXT RETRIEVAL ON FLICKR30K AND MS COCO. R@K CORRESPONDS TO WHETHER THE GROUND TRUTH IS RECALLED IN THE TOP-K RESULTS. ‘-LARGE’ DENOTES THE LARGE VERSION

Model	Flickr30k							MS COCO						
	Image Retrieval			Text Retrieval			Overall	Image Retrieval			Text Retrieval			Overall
	R@1	R@5	R@10	R@1	R@5	R@10	Sum	R@1	R@5	R@10	R@1	R@5	R@10	Sum
Uncicoder-VL-large [57]	73.60	92.30	95.80	86.50	97.60	99.30	545.10	-	-	-	-	-	-	-
VILBERT-large [51]	58.78	85.60	91.42	-	-	-	-	-	-	-	-	-	-	-
UNITER-large [53]	73.56	94.08	96.76	87.30	98.00	99.20	548.90	52.93	79.93	87.95	65.68	88.56	93.76	468.81
CSIC-large	75.32	93.60	96.78	88.50	98.10	99.40	551.70	53.20	80.12	88.09	67.46	89.26	94.32	472.45

and 1.20% (R@1) for text retrieval, for a total performance improvement of 2.80%.

On MS COCO, our approach also achieves better performance than baseline on two retrieval tasks. In particular, compared to the UNITER-base, our approach achieves maximum performance improvements of 0.64% (R@1) and 2.26% (R@1) in the two retrieve tasks, respectively, with a 5.55% improvement in the sum of metrics. Our method also shows significant improvement compared to UNITER-large. CSIC achieves an improvement of 0.27% (R@1), 1.78% (R@1) for image retrieval and text retrieval with a total performance improvement of 3.64%.

We can also find other details. The performance improvement of text retrieval is better than image retrieval for both the base and large models. This phenomenon is seen on both Flickr30k and MS COCO. We analyze that this is due to the fact that images have richer semantic information than text, and it is often more accurate to use the rich information to retrieve refined information. Also due to the dataset structure, each image is accompanied by 5 annotation texts, and our approach helps to achieve better image-text alignment, making it easier to obtain ground-truth for text retrieval. In addition, the base version shows a 9.51% performance improvement which is higher than the large. This suggests that the large version with more transformer layers can achieve better cross-modal semantic importance consistency than the base version.

D. Model Analysis

We carry out extensive experiments to verify the soundness of our approach and to explore the effect of parameter variation on the model effectiveness. The model analysis is conducted on Flickr30k with the base version of the proposed method.

1) *Ablation Study*: In previous experiments, we applied CSIC constraints to sequences of both visual and textual modalities and achieved significant improvements. In order to investigate the effects of different modalities on the model effects, we conducted ablation experiments and the results are listed in Table III. CSIC- $\mathbb{L}$ indicates that CSIC is added to the textual modality only, and CSIC- $\mathbb{V}$ indicates that CSIC is added to the visual modality only. CSIC can significantly improve model performance when applied to either visual or textual modalities, with performance improvement of %1.64 and %1.68, respectively. But applying CSIC in both modalities simultaneously achieves the best results which indicates that it is necessary to apply CSIC constraints in both

TABLE III
RESULTS OF ABLATION EXPERIMENTS ON FLICKR30K FOR THE UNITER BASE. ‘-L’ INDICATES THAT CSIC IS ADDED TO THE TEXTUAL MODALITY ONLY, AND SIMILAR FOR ‘-V’. ‘*’ DENOTES OUR REPRODUCTION RESULTS

Model	Image Retrieval			Text Retrieval		
	R@1	R@5	R@10	R@1	R@5	R@10
UNITER-base	72.52	92.36	96.08	85.90	97.10	98.80
UNITER-base*	72.26	92.36	95.98	85.90	97.40	98.90
CSIC-L	73.08	92.82	96.34	85.70	97.40	99.10
CSIC-V	72.86	92.80	96.14	86.10	97.30	99.20
CSIC	73.42	93.02	96.28	86.90	97.80	99.30

modalities. Although the sum values of the metrics improvement by introducing CSIC in each of the two modalities are almost equal, there is a bias in the detailed metrics. This is due to the fact that CSIC applied to text modality allows $\mathbf{S}_{\mathbb{L}\mathbb{L}}$ and $\mathbf{S}_{\mathbb{V}\mathbb{L}}$ to correct each other, and visual items can learn more reasonable context vectors under the influence of $\mathbf{S}_{\mathbb{V}\mathbb{L}}$, which allows the model to calculate more accurate text-to-image similarity scores. This is similar to the visual modality, and the best results are obtained by applying CSIC to both modalities. The performance improvement for the baseline is better than the sum of the results of applying it to both modalities separately, which is the result of the two modalities promoting each other.

2) *Effect of Annealing Schedule*: In addition to the log schedule of Eq.(15), we also use the following multiple schedules:

- log schedule: $\lambda_s = 1 - \exp(-s/S \times \tau)$;
- linear schedule: $\lambda_s = s/S$;
- exp schedule: $\lambda_s = \exp((s/S - 1) \times \tau)$,

where $\tau \in \{5, 10\}$. Visualize results of schedules are shown in Fig. 4. We compare five annealing schedules and the comparison results are reported in Table IV. Besides, the sum of the metrics is presented in Fig. 5 to visualize the impact of different schedules on the model performance. According to the experimental results, it can be seen that our method can obtain a more significant performance improvement regardless of the strategy. This indicates the good robustness of the proposed method. The best model performance is achieved with log schedules at $\tau = 5$, which outperforms other schedules in most metrics.

3) *Effect of Layer to Apply CSIC*: Self-attention achieves the fusion of individual modal information by computing the

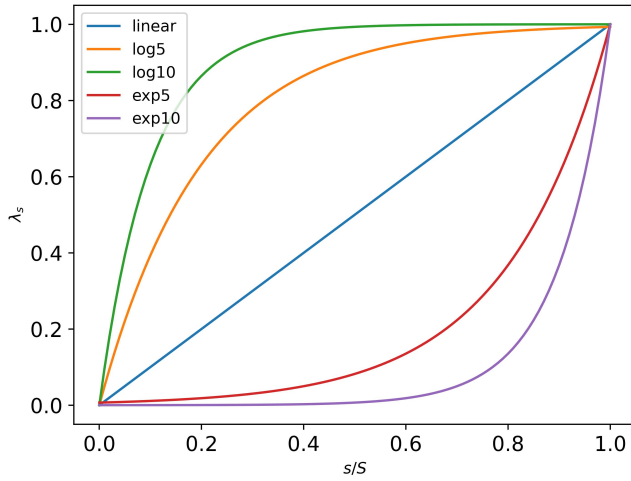
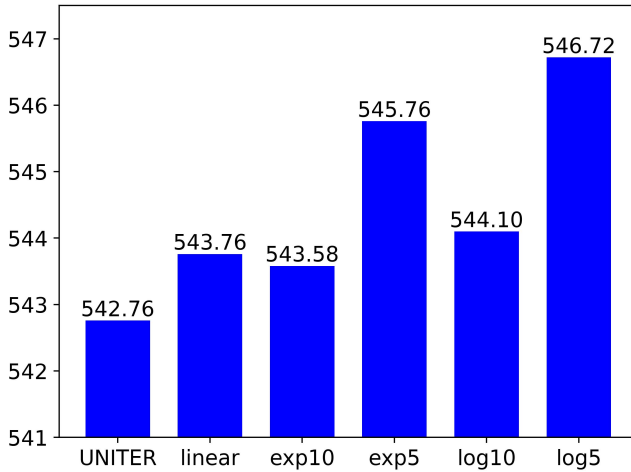


Fig. 4. Signal annealing change curve.

TABLE IV
RESULTS FOR DIFFERENT ANNEALING SCHEDULE, ‘*’ DENOTES
OUR REPRODUCTION RESULTS

Model	Image Retrieval			Text Retrieval		
	R@1	R@5	R@10	R@1	R@5	R@10
UNITER-base	72.52	92.36	96.08	85.90	97.10	98.80
UNITER-base*	72.26	92.36	95.98	85.90	97.40	98.90
+ linear	72.74	92.60	96.12	85.30	97.90	99.10
+ exp10	71.90	92.48	96.10	87.00	97.40	98.70
+ exp5	72.86	92.80	96.14	86.10	97.30	99.20
+ log10	73.30	92.76	96.24	85.50	97.50	98.80
+ log5	73.42	93.02	96.28	86.90	97.80	99.30

Fig. 5. ‘Sum’ for different training signal annealing. The proposed method achieves the best results when the log schedule with $\tau = 5$ is used for the model effect.

similarity of the representation vectors of each input item, and also aligns the modal information. However, single-layer self-attention does not fuse the multimodal information well. When multiple layers of self-attention are superimposed, the abstraction ability of the model increases, which helps the model to explore the correspondence between different modal

TABLE V
RESULTS OF DIFFERENT LAYERS TO APPLY CSIC. ‘*’ DENOTES OUR
REPRODUCTION RESULTS

Model	Image Retrieval			Text Retrieval		
	R@1	R@5	R@10	R@1	R@5	R@10
UNITER-base	72.52	92.36	96.08	85.90	97.10	98.80
UNITER-base*	72.26	92.36	95.98	85.90	97.40	98.90
layer4	72.18	92.42	96.06	86.40	97.80	98.80
layer8	72.56	92.32	96.24	86.80	97.40	99.10
layer12	73.42	93.02	96.28	86.90	97.80	99.30

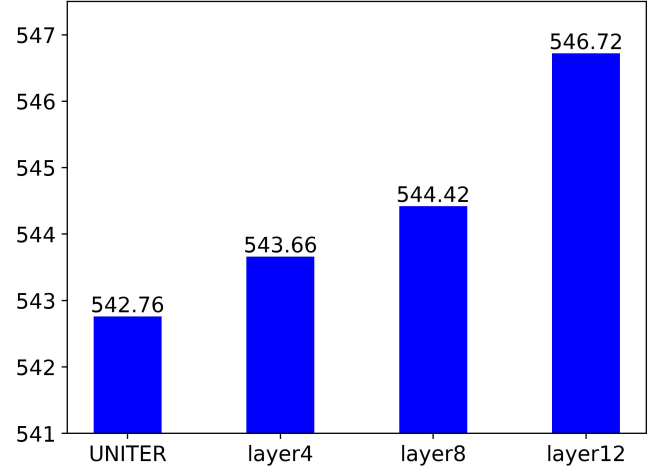


Fig. 6. Applying CSIC on different layers. The performance metrics get higher as the number of CSIC application layer increases.

information. To explore the effect of the number of self-attention layers, we also apply CSIC on different self-attention layers, and the results are listed in Table V, while the sum of metrics is shown in Fig. 6. The experimental results show that our method has a more significant improvement compared to UNITER regardless of the layer where CSIC is applied. Here only a few layers are selected for comparison in equal order of difference. We can see that the model achieves the best results when CSIC is applied on the highest layer. Also we find that the improvement of CSIC on the model effect is more obvious as the number of applied layers increases. We analyze that because higher layers of the transformer allow for better image-text alignment, and the proposed CSIC is a further strong constraint in the ideal alignment state. Also, the representation vector at higher layers has more influence on the final output of the model, so when CSIC is applied at the highest layer, it helps to improve the model performance more.

E. Qualitative Results

Fig. 7 shows some examples of visualization of cross-modal semantic importance changes of image regions and text words. Each example selects two regions of the target in the image-text and the corresponding words. To compare the differences more intuitively, we calculate and show their intra-modal and inter-modal semantic importance differences. We compare the degree of semantic importance consistency

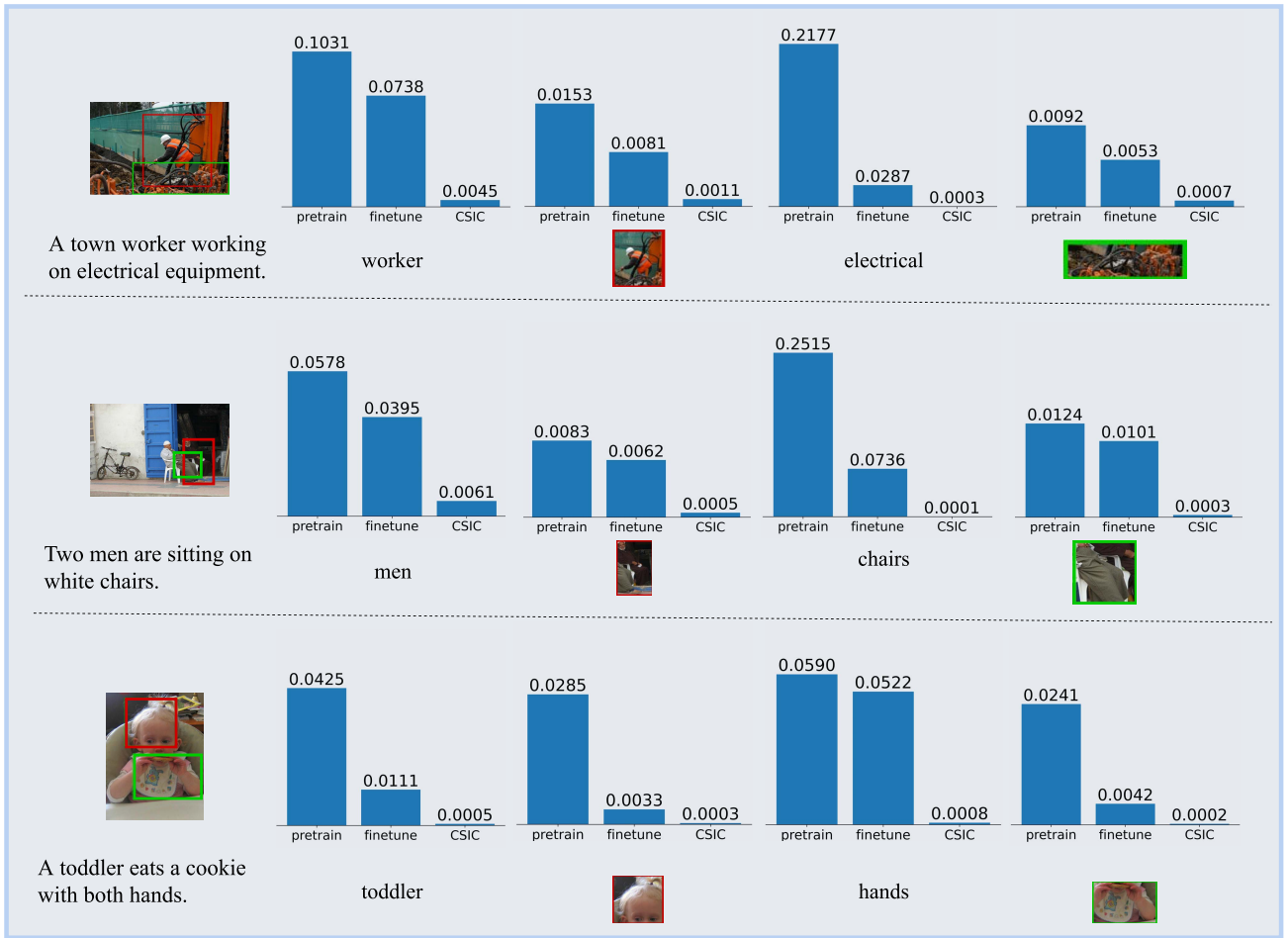


Fig. 7. Visualization examples of cross-modal semantic importance in different methods on Flickr30k. The semantic importance weights of four items are listed in an image-text pair, and the bar represents the intra-modal and inter-modal semantic importance differences of the items.



Fig. 8. Examples of text retrieval results by CSIC, UNITER [53], SCAN [24] and IMRAM [37]. The texts marked in green are correct, while those marked in red are incorrect, conversely.

on the three methods, i.e., pretrain, finetune, and CSIC, which denote UNITER pre-trained model, UNITER fine-tuned model, and the proposed method, respectively. We can see that the semantic importance of items is converging based



Fig. 9. Examples of image retrieval results by CSIC, UNITER [53], SCAN [24] and IMRAM [37]. We only show top-3 similar results and those marked in green box is the ground truth image, others marked in red box are incorrect.

on the above order, which can verify the rationality of our method. Compared with UNITER-pretrain, CSIC can enhance the semantic importance agreement of regions and words, and also makes the semantic importance distribution of items converge to a reasonable level, so our method can obtain further performance improvement.

Fig. 8 and Fig.9 show some examples of text retrieval and image retrieval, respectively, comparing the proposed method with other baselines, and the correct results are marked in green. We can see that CSIC can accommodate more query samples and thus improve the recall of retrieval. For the text retrieval task, CSIC recalls the most ground truth texts in the three samples. In the example of Fig. 8(a), the UNITER does not retrieve a single correct result under the condition that there are 5 ground truth texts for an image. In contrast, CSIC successfully retrieves 3 ground truth texts. For the samples of image retrieval, CSIC achieves the recall of top 1. In contrast, other baselines either do not recall the ground truth or the ground truth is poorly ranked.

V. CONCLUSION

In this paper, we propose Cross-modal Semantic Importance Consistency (CSIC) for cross-modal image-text retrieval. CSIC extracts the semantic features at the object level using Faster R-CNN and tokenizes the text. Then the intra-modal and inter-modal semantic importance of items is measured based multimodal attention matrix. Then, during the training process, the two types of importance are gradually consistent. The CSIC can correct some cases where the semantic importance of items is not reasonable, allowing the model to learn more accurate representation vectors for items. We carried out an extensive experimental analysis on two benchmark datasets to verify the rationality and effectiveness of our method. Quantitative results show that CSIC achieves significant improvement in both image retrieval and text retrieval. On the other hand, qualitative results show that CSIC has good robustness and enables better image-text alignment. We note that ALBEF [64] outperforms CSIC for image-text retrieval. The reason is that compared to UNITER, ALBEF uses a stronger feature encoder and aligning before cross-modal fusion, which results

in a much better performance than UNITER. In future research work, we will explore the application of the proposed approach to non-region-based models such as ALBEF. Encoding images by vision transformer, there will be more non-significant features for cross-modal interaction, which is a notable challenge for semantic importance metrics.

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